



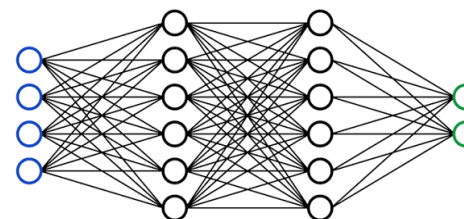
**UNIVERSITÀ
DEGLI STUDI
DI TRIESTE**

Dipartimento di
**Matematica, Informatica
e Geoscienze**

HYBRIDIZING DEEP LEARNING AND CLASSICAL MATHEMATICAL MODELING

THEORY AND NEW APPLICATIONS

$$\dot{x}(t) = f(x(t), t,$$



)

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INTRODUCTION: HYBRID MODELING

MACHINE LEARNING

Learn patterns from **data**

MATHEMATICAL MODELING

Derive from **governing laws**

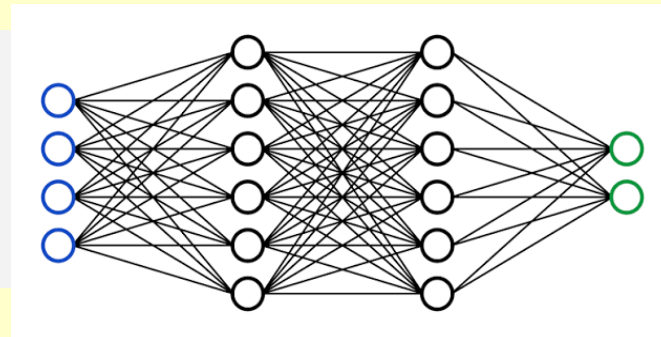
HYBRID MODELING

Data + Physics-based Model

Our Case Study: Universal Differential Equations

- C. Rackauckas (MIT), *Universal Differential Equations for Scientific Machine Learning* (2020)
- **Intuition:** Neural Networks inside Differential Equations!

$$\dot{x}(t) = f(x(t), t,$$



)

METHODOLOGY: A GENERAL IDEA

Sketch of the methodology used in the experiments

Known Model
(ODE)

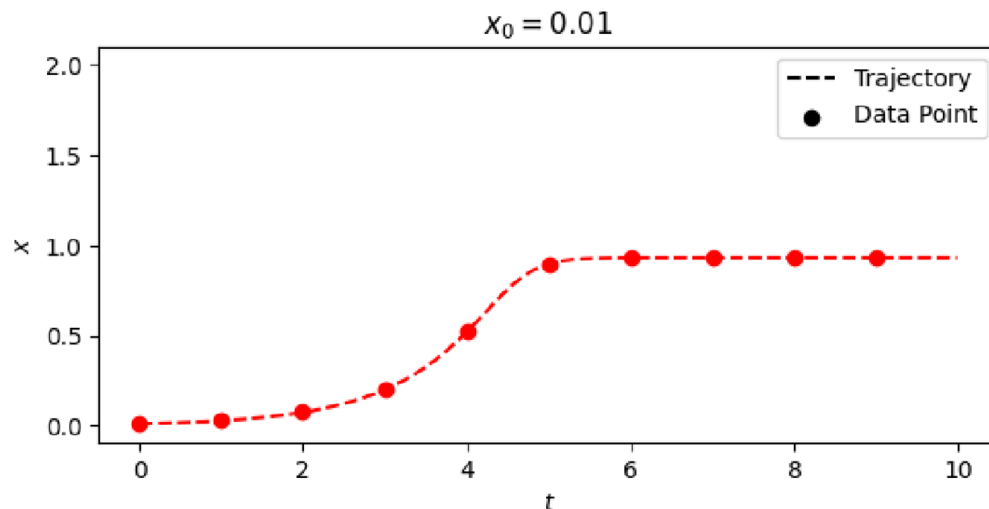
$$\begin{aligned}\dot{x} &= x(\alpha - \beta y) \\ \dot{y} &= y(\gamma x - \delta)\end{aligned}$$

Hybrid Model
(UDE)

«Observed» trajectory (time series)

Train on

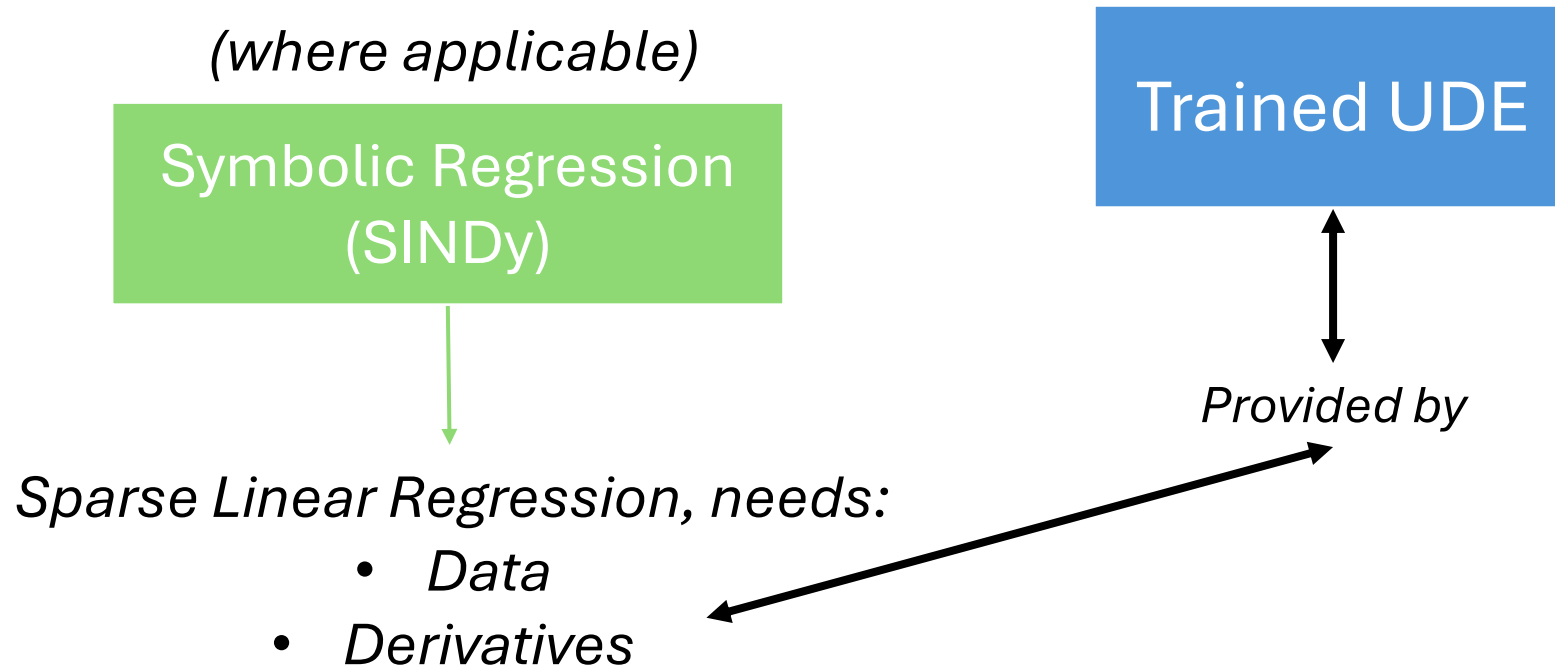
«Replace» unknown parts
with Neural Networks



$$\begin{aligned}\dot{x} &= \alpha x - U^{(1)}(x, y; \theta) \\ \dot{y} &= -\beta y + U^{(2)}(x, y; \theta)\end{aligned}$$

METHODOLOGY: A GENERAL IDEA

Sketch of the methodology used in the experiments



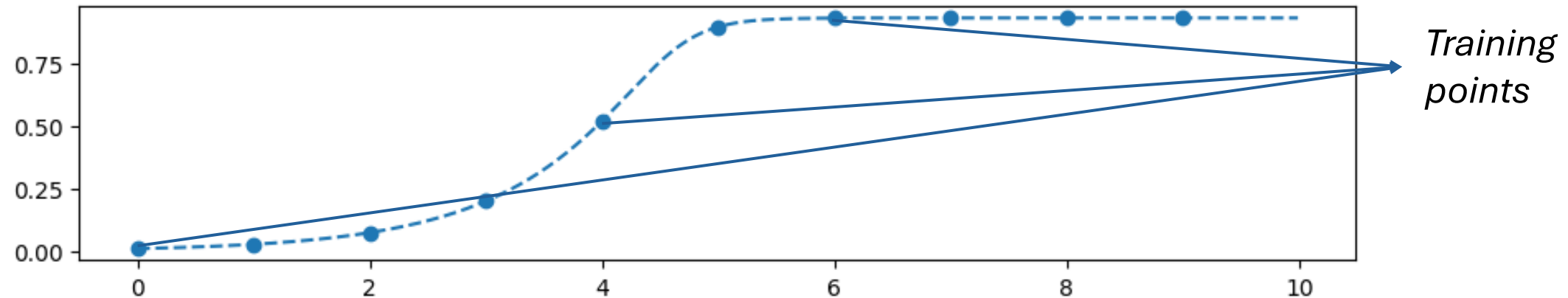
GENERALIZED LOGISTIC EQUATION

Modeling a population's growth

$$\dot{N} = rN(1 - \rho_1 N - \rho_2 N^2 - \dots - \rho_k N^k)$$

Imagine we lost the polynomial but we still have a part of time series

$$\dot{N} = rN(1 - \text{?})$$



How do we recover it with a small sampling of the time series?

FIRST CASE STUDY: LOGISTIC EQUATION

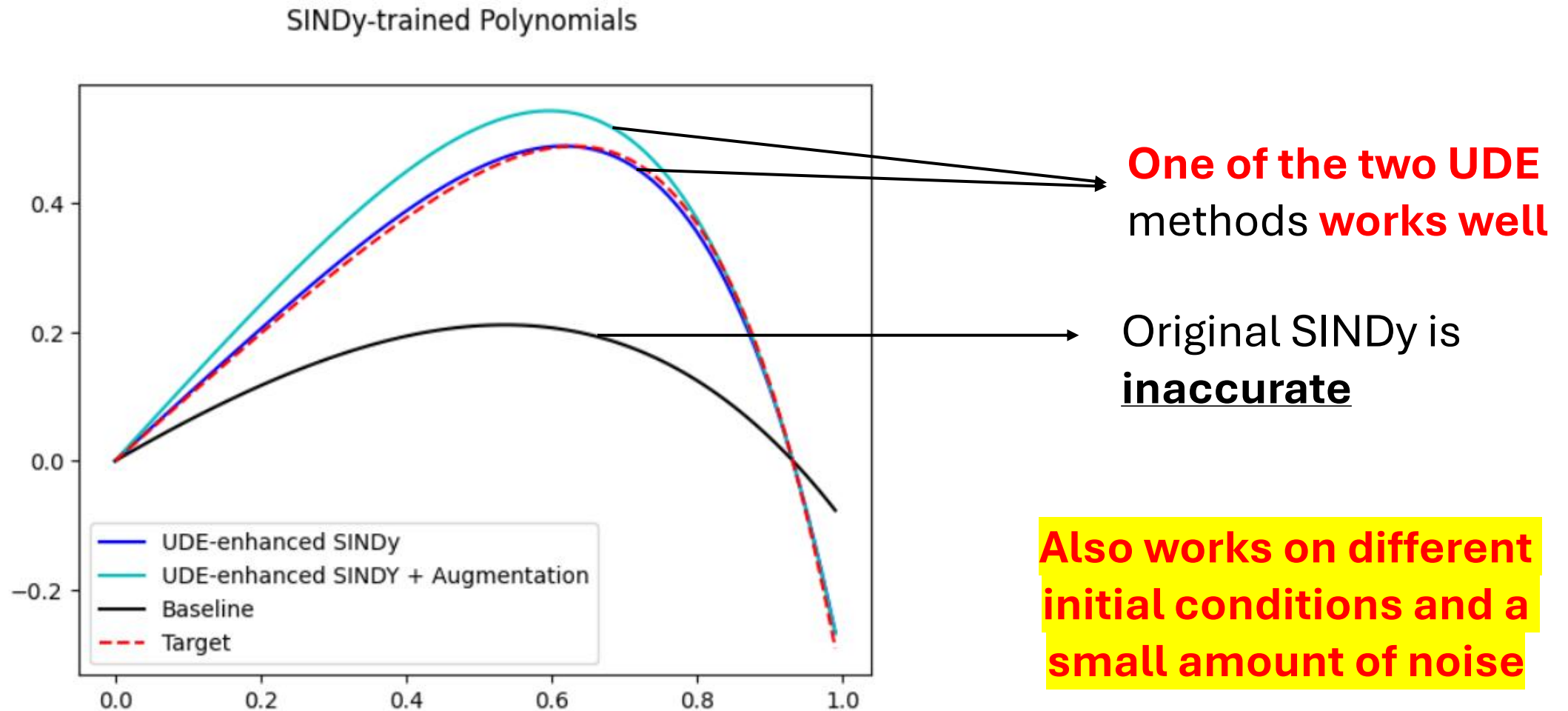


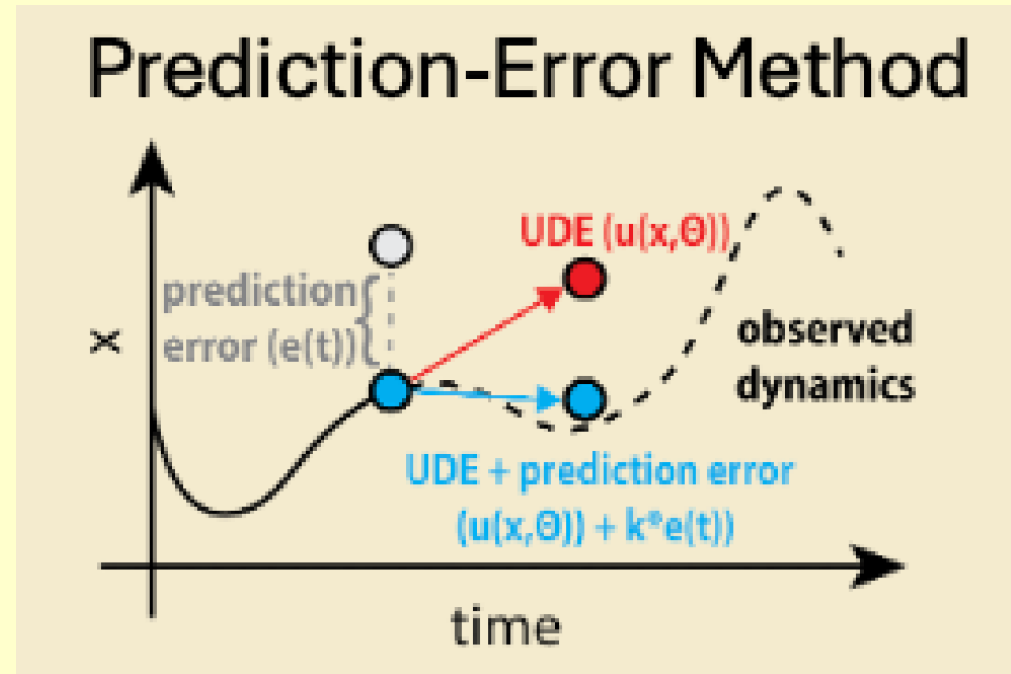
Figure 3.1.2: Generalized Logistic First Experiment Results (SINDy)

PEM-UDEs: A STATE OF ART VARIANT

Proposed recently (<1 year) by Anthony G. Chesebro

- Add *prediction-error correction* term
- Improves training on **Chaotic** Systems

$$\frac{dx}{dt}(t) = F(t, x) + \overset{\text{The «new» part}}{K\varepsilon(t)} + U(x; \theta)$$
$$\varepsilon(t) = \hat{x}(t) - x(t)$$



A CHAOTIC CASE STUDY: CHUA'S CIRCUIT

Chua's Circuit: very important and very studied

- **Why? Since First example of measured chaos!** (oscilloscopes)
- Chaotic Motion with a strange attractor oscillator

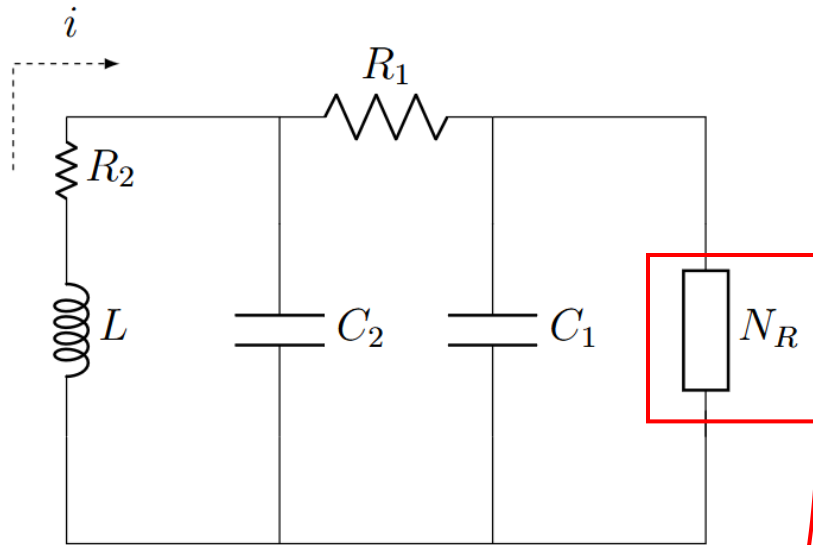
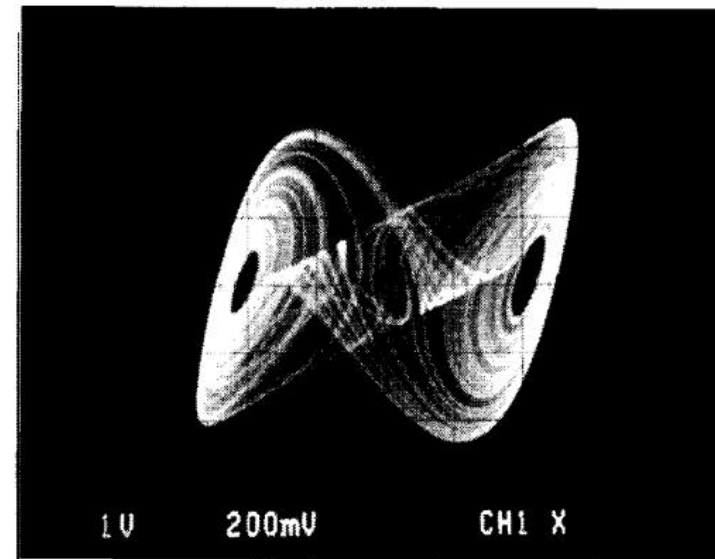


Figure 1.5.16: Chua's Circuit as in [4]

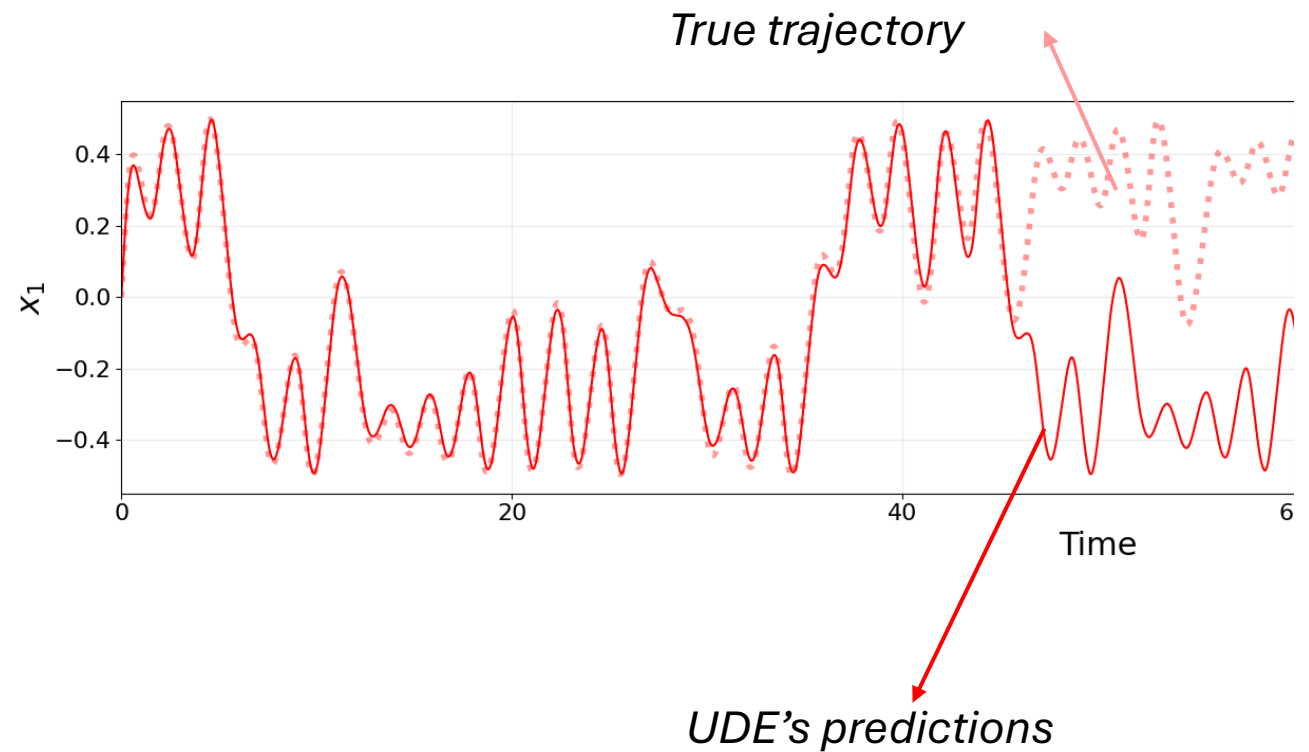
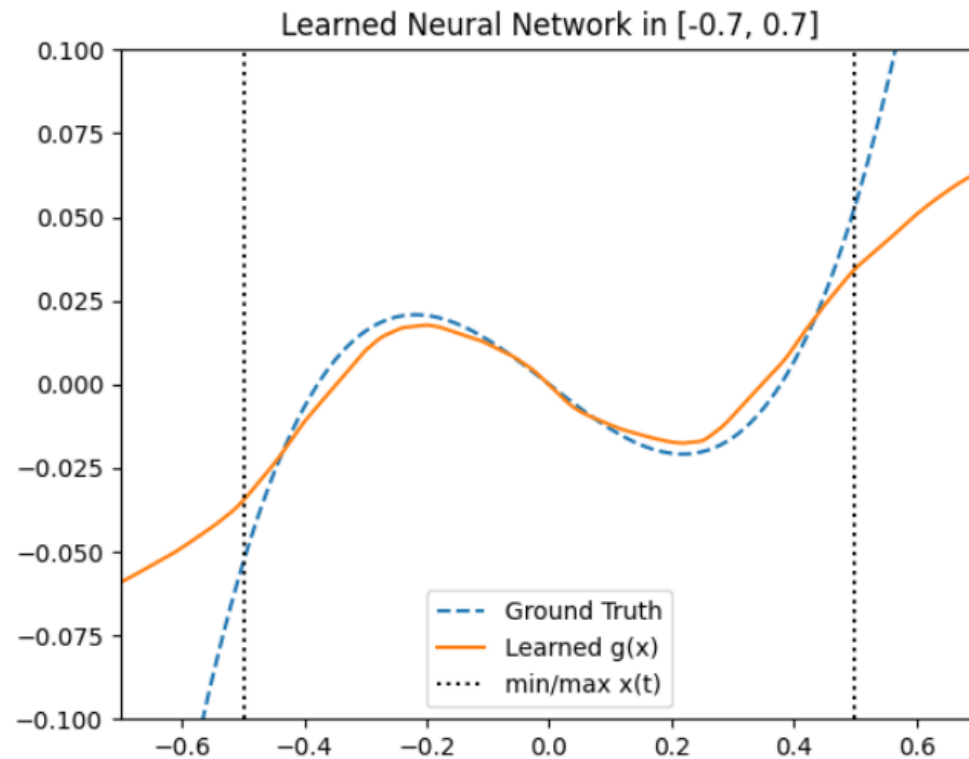


(p)

Can we replicate **Chua's term** with a sample of the data?

A CHAOTIC CASE STUDY: CHUA'S CIRCUIT

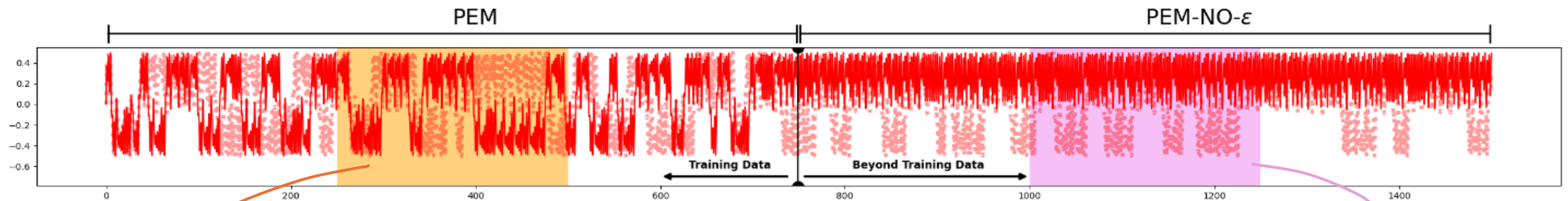
- Learned nonlinearity matches well ground truth
- Prediction diverges over time (expected in chaotic systems)



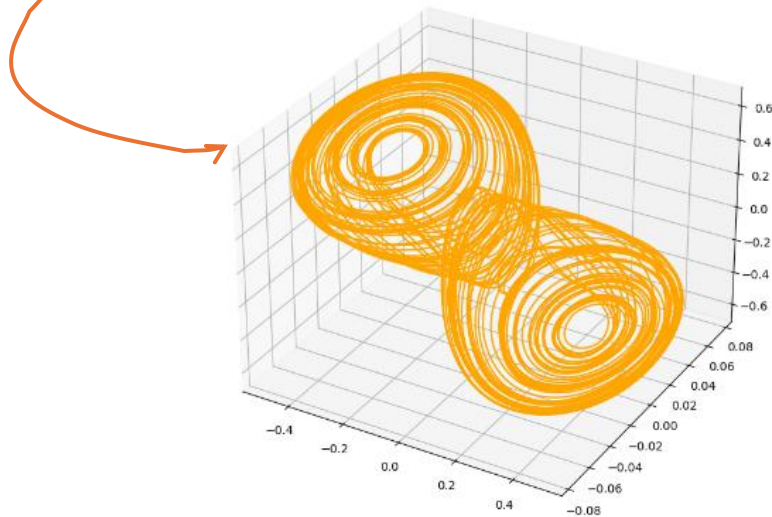
CHANGE OF STRANGE ATTRACTOR!!!

- We are writing a **scientific manuscript** on these chaos-related results and other (e.g. transient chaos in the hybrid model)

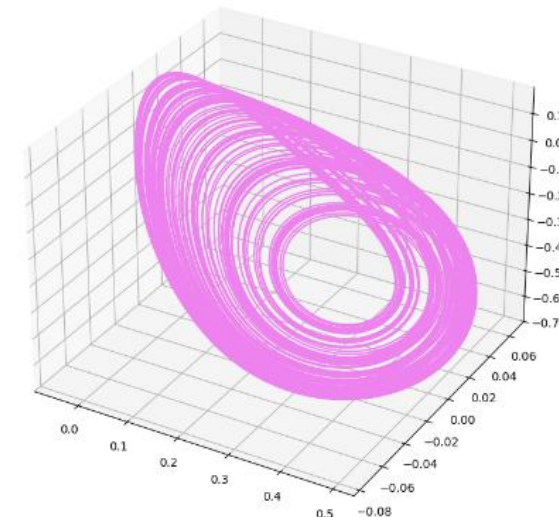
(a) Time Series for $0 \leq t \leq 1500$



(b) 3D plot of trajectory for $250 \leq t \leq 500$



(c) 3D plot of trajectory for $1000 \leq t \leq 1250$



A NEW CASE STUDY: NEURONAL MODELING

Neural Mass Model: Modeling biological and **future artificial neurons**

$$\tau \dot{r} = -\boxed{r} + \boxed{\Phi}(\boxed{I_{\text{ext}}} + Jr)$$

«Firing Rate»
Neuron's Activity

«Activation Function»
Input into Firing Rate
Usually depends on a
threshold value

«External Impulse»
A signal from outside

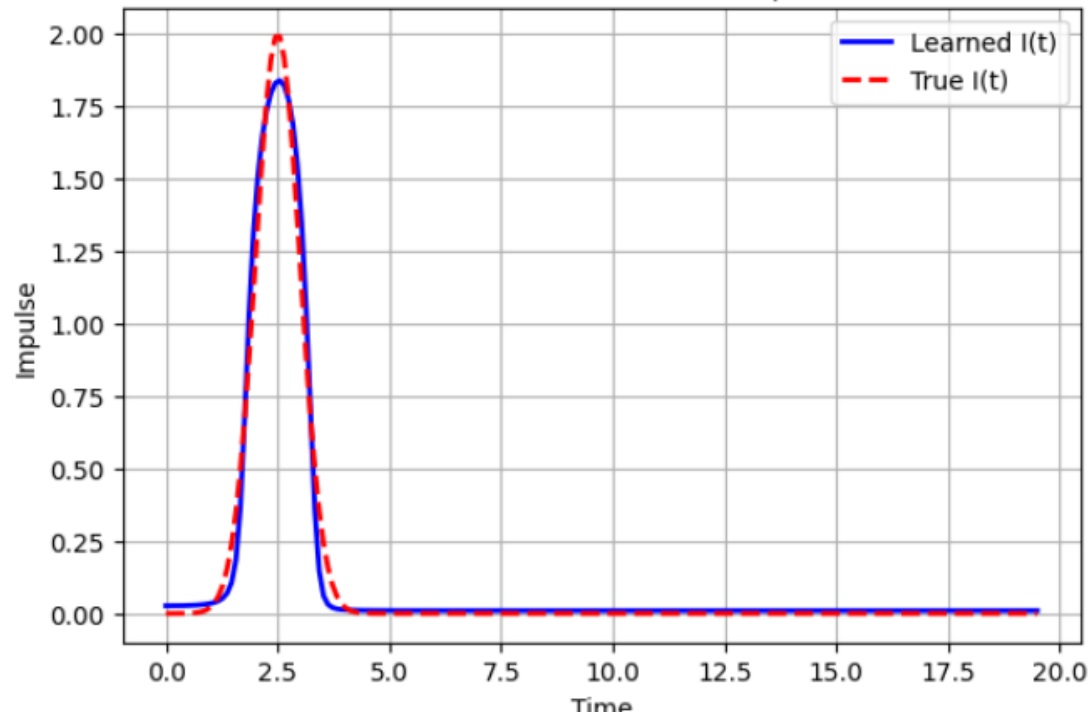
Can we learn the components of our neuron, such as the impulse, with UDEs?

$$\tau \dot{r} = -r + \Phi(\boxed{?} + Jr)$$

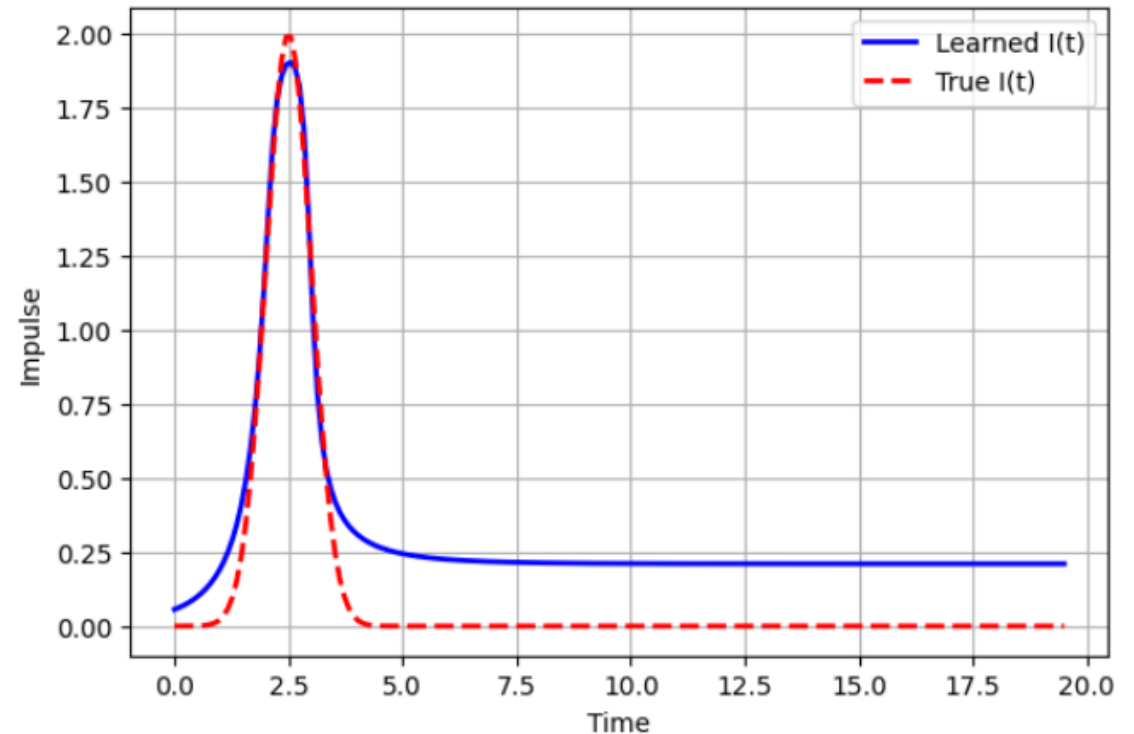
A NEW CASE STUDY: NEURONAL MODELING

- Main features of the impulse are recovered (*the «spike»*)
- The exact shape depends on the identifiability of the system

Case 1: Low Threshold Value



Case 2: High Threshold Value



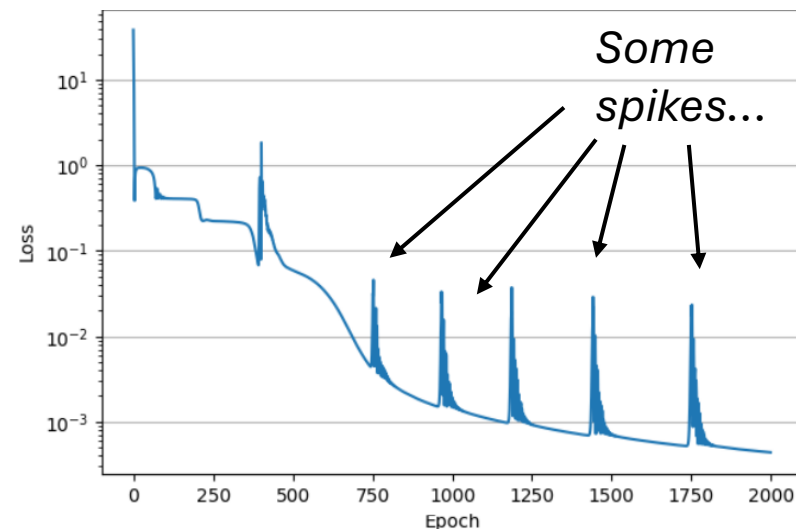
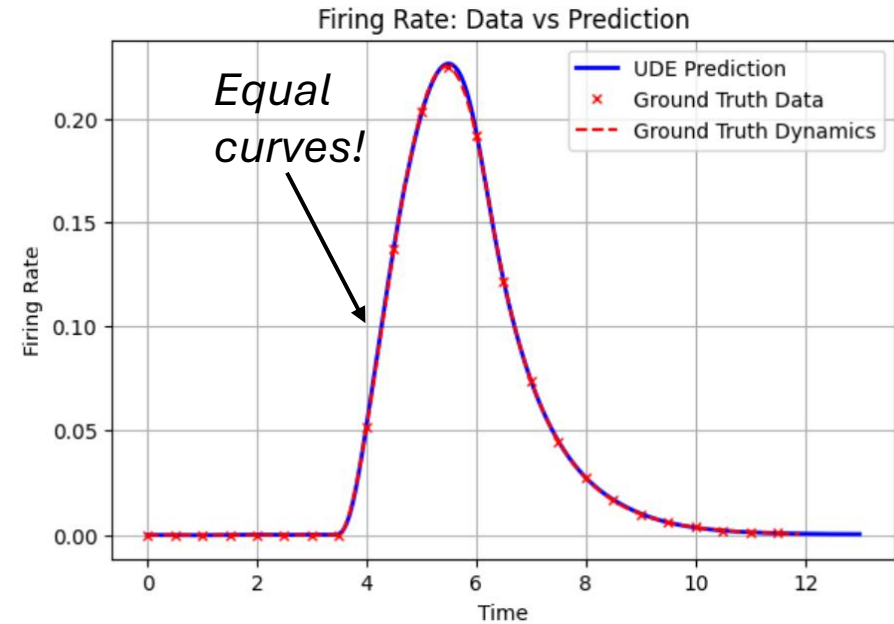
Recap + Other aspects

Main findings on UDEs

- Generally work «well enough»
 - Identify missing dynamics
 - Recover symbolic equations with SINDy
- Strong interpolation capabilities

Some challenges of UDEs

- Model identifiability remains challenging
- Instability during training



FUTURE WORK

Q: What can we do further?

Other fields of application: E.g. Physics, Epidemiology, Chemistry, Materials Science, Biology, et cetera...!

Beyond ODEs: SDEs, PDEs, IDEs, ...

$$dX_t = \mu dt + \sigma dB_t$$

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$$

$$\tau \frac{dr}{dt}(x, t) = -r(x, t) + \Phi \left(I_{\text{ext}} + \int_{\Omega} J(x, y) r(y, t) dy \right)$$

A vast field of potentialities!

Thanks for listening!

“Essentially, all models are wrong, but some are useful”

**G. Box, “The Use of Approximating
Functions” (1976)**